

I design language platforms and runtimes. I created UniversalToolchain/Wist2, a modular .NET SDK with typed package composition, exact manifest binding, callable-first SSA, interpreter/CIL execution, and experimental PlanFuzz.

Platform architecture

Typed contracts, independent packages, capabilities and conflicts, deterministic routes

Verifiable semantics

Callable-first SSA, call effects and trust, verifier-gated AIR → SSA → AIR

1,465 / 1,465

Zero failures across nine packages, clean consumers, and interpreter/CIL parity

EXPERIENCE		SKILLS
2024 — present	UniversalToolchain / Wist2 — creator and primary developer - Defined package boundaries, artifact contracts, capability/conflict resolution, routing, and pass ordering. - Implemented exact manifest binding, lock identity, runtime-session lifecycle ownership, and fail-closed policies.	Language architecture typed artifacts, package graphs, capabilities/conflicts, routes, passes, manifests, and lock identity IR / SSA Bytecode/AIR, callable-first SSA, CFG, dominance, folding, SCCP-lite, unreachable elimination, DCE Runtime interpreter/CIL, typed delegates, lifecycle ownership, fail-closed policies, backend parity Verification differential/metamorphic testing, exact oracles, replay, reducers, sanitizers, clean consumers
July 1 — August 31, 2026	MCST — compiler engineering intern Developing an LLVM optimization pass and testing graph algorithms.	
2023 — 2025	Mama Lama — Software Developer Independently designed the architecture and modular structure of event applications and took them from requirements to finished products.	
PROJECTS		EDUCATION
UniversalToolchain / Wist2 1,465/1,465 tests with zero failures across nine packages and clean-consumer projects, plus interpreter/CIL parity and reproducible configuration checks.		Software Engineering student at HSE University
PlanFuzz Reproducible artifacts and explicit separation of confirmed, unstable, inconclusive, and infrastructure outcomes.		RECOGNITION HSE Vysshaya Proba prize-winner in competitive and industrial programming; regional Russian informatics olympiad prize-winner.
PS-form Memory Dependence Analyzer Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples.		Moscow / remote